Nicholas Batchelder

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Education

University of Washington — Seattle, WA
Masters of Science in Computer Science — *Autumn 2025 - Winter 2026*

University of Washington — Seattle, WA
Bachelor of Science in Computer Science, Minor in Linguistics; GPA: 3.84 — *Autumn 2022 - Spring 2025*

Work Experience

Software Engineer — San Mateo, CA
Roblox Corporation — *Starting August 2026*

Part-Time Android Developer — Seattle, WA
AppEsteem Corporation — *January 2026-April 2026*

Software Engineering Intern — San Mateo, CA
Roblox Corporation — *May 2025 – August 2025*

- Developing software to identify and visualize DDOS attacks on Roblox games in real-time, providing actionable insights to creators about when and how their experiences are targeted and successfully defended.
- Designing predictive systems that analyze attack patterns to support Roblox in capacity and defense planning.

Part-Time Backend Engineer — Seattle, WA
AppEsteem Corporation — *January 2025-May 2025*

- Developed a .NET-based malware scanner leveraging AppEsteem's Deceptor Database for a Microsoft-backed project to automatically detect potentially unwanted software in Microsoft App Store submissions.

Research Assistant — Seattle, WA
Mobile Intelligence Lab — *June 2024 – Present*

- Designed LlamaPIE: a custom architecture LLM "proactive" conversational speech agent that analyzes human speech in real-time and provides unprompted advice to its user when helpful (paper link here).
- Created and deployed an LLM-based dialogue creation pipeline, allowing further conversational LLM research to be conducted using customized, task-specific synthetic dialogues.

AI Development Intern — Seattle, WA
Three Clicks Corporation — *June 2023 – September 2023*

- Developed a proof-of-concept "app crawler" to automatically install applications by navigating through installation windows using an LLM, Text Recognition Model, and Object Detection Model.
- Implemented restraints to prevent "hallucinating" button responses or clicking the wrong button by tracking screen data with a LanceDB vector database and limiting repeat button clicks using graphing algorithms.

Projects

Realtime Doodle Inpainting — University of Washington
Deep Learning — *Winter 2024*

- Developed a branched auto-regressive decoder to complete sketches in real-time using the Mamba Architecture.
- Optimized drawing and completion performance to real-time latency on User-Hardware (Integrated Graphics).
- Repo and Paper: github.com/eashvere/realtime-drawing-mamba

xk Operating System — University of Washington
Operating Systems — *Fall 2023*

- Developed the Operating System's file system, pipe, copy-on-write fork, and exec capabilities.

Patent — US 11,507,269 B2
Technologies for Indicating Third Party Content and Resources on Mobile Devices — *November 2022*

Skills

Languages: Python, C/C++, C#, Java, TypeScript, Lua, HTML/CSS
Technologies: Jetpack Compose, Docker, Pytorch, Langchain, Numpy, Git, Vim, Linux, React, Node.js, Svelte